

University of Wisconsin-Milwaukee

Subject: - INFOST 582G - Introduction to Data Science

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Assignment: - Final Project

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Habitat Fragmentation Pattern Analysis for Threatened and Endangered Species

I. Executive Summary:

This study analyzes the habitat fragmentation patterns across a selection of 9 different species from the FWS Critical Habitat Dataset. I used spatial visualization and geographic analysis to understand the variation in severity of fragmentation based on geographical distribution, taxonomic classification, and physiological characteristics. The key findings reveal that fishes have distinct fragmentation patterns and species endemic to islands show their own unique fragmentation. Mainland species that are highly specialized in a specific environment show intense fragmentation and are usually distributed far apart. Temporal analysis gives insight into how the size of the fragment has shrunk overtime and how the later designated species show more severe fragmentation. Through this analysis I will provide valuable insights into habitat conservation and help in identifying the most vulnerable species.

II. Introduction:

Habitat fragmentation is one of the greatest threats to biodiversity conservation and ecology on a global scale. If a fragmented species is not helped in time, it will lead to irreversible damage to the ecosystem, which in a worst-case scenario can result in a domino effect. The designation of critical habitat under the US Endangered Species Act of 1973 gives me an opportunity to look at fragmentation across the different species and geographical regions of the United States of America.

The fragmentation of once large continuous territories into disconnected patches has affected the species differently based on their mobility, territorial requirements, seasonal migration needs and ecological specialization. In this project/study I will examine the habitats of 10 selected species and analyze their critical habitat designations. With the help of Google Earth Pro and several spatial analysis techniques I will identify the fragmentation patterns, compare the degree of fragmentation among the selected species and assess the possible connectivity challenges.

III. Background and Questions:

The Critical habitat designation is put in place to safeguard areas that are essential to survival and conservation of endangered and endemic species. The effectiveness of these designations depends mainly on the number, size, and connectivity of the known fragments.

In this project I will address the following:

1. How do fragmentation patterns differ amongst different types of species and different types of habitats?
2. Do islandic and aquatic species show any unique form of fragmentation?
3. How does the size, number and how far apart the fragments are, affect a species?
4. How do regional environmental factors influence fragmentation?

IV. Data Collection:

From the primary source FWS Critical Habitat Dataset, I have downloaded all the files and selected the csv (converted to .xlsx for convenience) file “species_shapefile_map_20250328.xlsx” which has a total of 847 entries.

From which I have selected 10 different species for the scope of my analysis.

They are:

1. Buena Vista Lake ornate shrew (Sorex ornatus relictus): Mammal
2. Bull Trout (Salvelinus confluentus): Fish
3. Canada Lynx (Lynx canadensis): Mammal
4. California condor (Gymnogyps californianus): Bird
5. California red-legged frog (Rana draytonii): Amphibian
6. Colorado pikeminnow (Ptychocheilus lucius): Fish
7. Desert tortoise (Gopherus agassizii): Reptile
8. Florida bonneted bat (Eumops floridanus): Mammal
9. Gray wolf (Canis lupus): Mammal
10. Haha (Cyanea Superba): A Hawaiian plant.

V. Data Structure:

Column1	Column2	Column3	Column4	Column5	Column6	Column7
comname	sciname	spcode	vipcode	population_abbreviation	population_description	shapefile
1. Buena Vista Lake ornate shrew (Sorex ornatus relictus)	Sorex ornatus relictus	0001	P01	Wherever found	Wherever found	FCI_Lakeview_alongshore_20000702.ap
2. Bull Trout (Salvelinus confluentus)	Salvelinus confluentus	0002	P01	Wherever found	Wherever found	FCI_Salmontrout_salmontrout_salmontrout_20000808.ap
3. Canada Lynx (Lynx canadensis)	Lynx canadensis	0003	P01	Wherever found	Wherever found	FCI_Lynxlynx_alongshore_20000702.ap
4. California condor (Gymnogyps californianus)	Gymnogyps californianus	0004	P01	Wherever found	Wherever found	FCI_Lynxlynx_alongshore_20000702.ap
5. California red-legged frog (Rana draytonii)	Rana draytonii	0005	P01	Wherever found	Wherever found	FCI_Lynxlynx_alongshore_20000702.ap
6. Colorado pikeminnow (Ptychocheilus lucius)	Ptychocheilus lucius	0006	P01	Wherever found	Wherever found	FCI_Lakeview_alongshore_20000702.ap
7. Desert tortoise (Gopherus agassizii)	Gopherus agassizii	0007	P01	Wherever found	Wherever found	FCI_Lakeview_alongshore_20000702.ap
8. Florida bonneted bat (Eumops floridanus)	Eumops floridanus	0008	P01	Wherever found	Wherever found	FCI_Lakeview_alongshore_20000702.ap
9. Gray wolf (Canis lupus)	Canis lupus	0009	P01	Wherever found	Wherever found	FCI_Lakeview_alongshore_20000702.ap
10. Haha (Cyanea Superba)	Cyanea Superba	0010	P01	Wherever found	Wherever found	FCI_Lakeview_alongshore_20000702.ap

The data structure in the spreadsheet has the following columns:

1. comname
2. sciname
3. spcode
4. vipcode
5. population_abbreviation

6. population_description

7. shapefile

VI. Data Cleaning:

1. Using text filter on the spcode column, all entries are filtered by species type.

2. The 10 species are selected, and their data is copied onto a separate sheet to safely work on them.

3. All associated the shape files are converted into KML files to use them with Google Earth Pro.

VII. Data analysis:

1. In the sperate worksheet named “Selected_Species”, new columns named ‘Geographic Region’, ‘Habitat Type’, ‘number of clusters’, ‘area of largest fragment’, ‘area of smallest fragment’, and ‘max separation’ are created.

Species	Spcode	Population	Geographic Region	Habitat Type	Number of Clusters	Area of Largest Fragment	Area of Smallest Fragment	Max Separation
Hyla	0104	100	Florida	Temperate Forest	4	100 sqm	10 sqm	100m
Pseudoeurycea	0105	100	Florida	Temperate Forest	4	100 sqm	10 sqm	100m
Dendrobates	0106	100	Florida	Temperate Forest	4	100 sqm	10 sqm	100m
Dendrobates	0107	100	Florida	Temperate Forest	4	100 sqm	10 sqm	100m
Dendrobates	0108	100	Florida	Temperate Forest	4	100 sqm	10 sqm	100m
Dendrobates	0109	100	Florida	Temperate Forest	4	100 sqm	10 sqm	100m
Dendrobates	0110	100	Florida	Temperate Forest	4	100 sqm	10 sqm	100m
Dendrobates	0111	100	Florida	Temperate Forest	4	100 sqm	10 sqm	100m
Dendrobates	0112	100	Florida	Temperate Forest	4	100 sqm	10 sqm	100m
Dendrobates	0113	100	Florida	Temperate Forest	4	100 sqm	10 sqm	100m

2. Using the KML files I obtained:

- Size of largest fragment
- Size of smallest fragment
- Number of clusters
- Maximum separation distance.

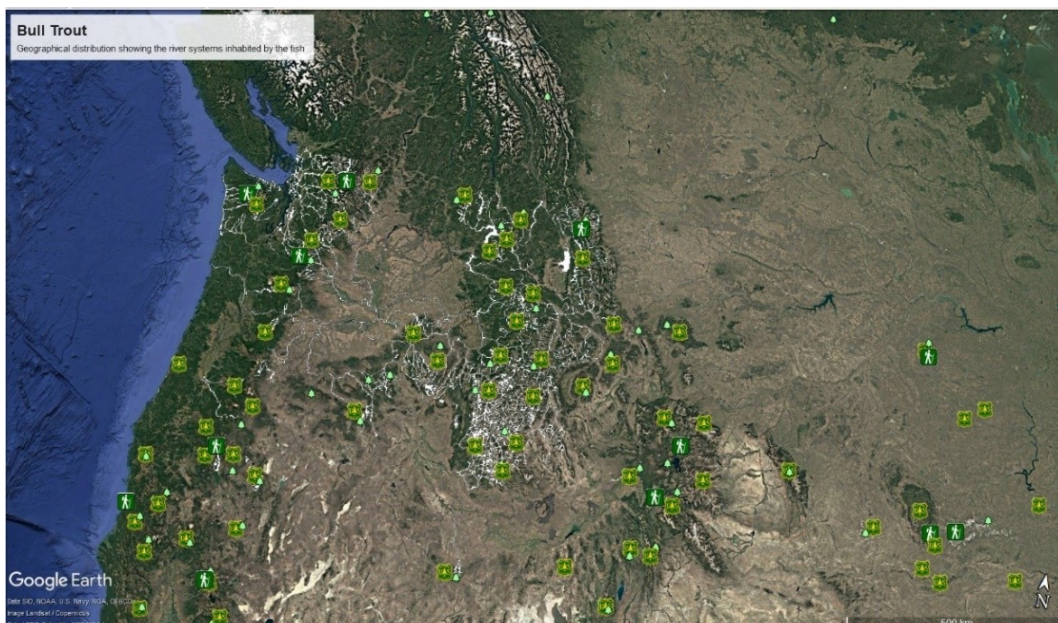
3. Analysis of each of the selected species:

3.1. Buena Vista Lake Ornate Shrew (*Sorex ornatus relictus*): Mammal



- i) This species exhibits the most fragmentation of all the selected species with just 4 tiny clusters. The largest patch is ~4.5 sq.km and smallest is ~0.4 sq.km.
- ii) Most fragments are separated by 20-45 km and the longest distance between fragments is ~124 km.
- iii) The fragments are almost completely isolated, which limits the mobility severely.

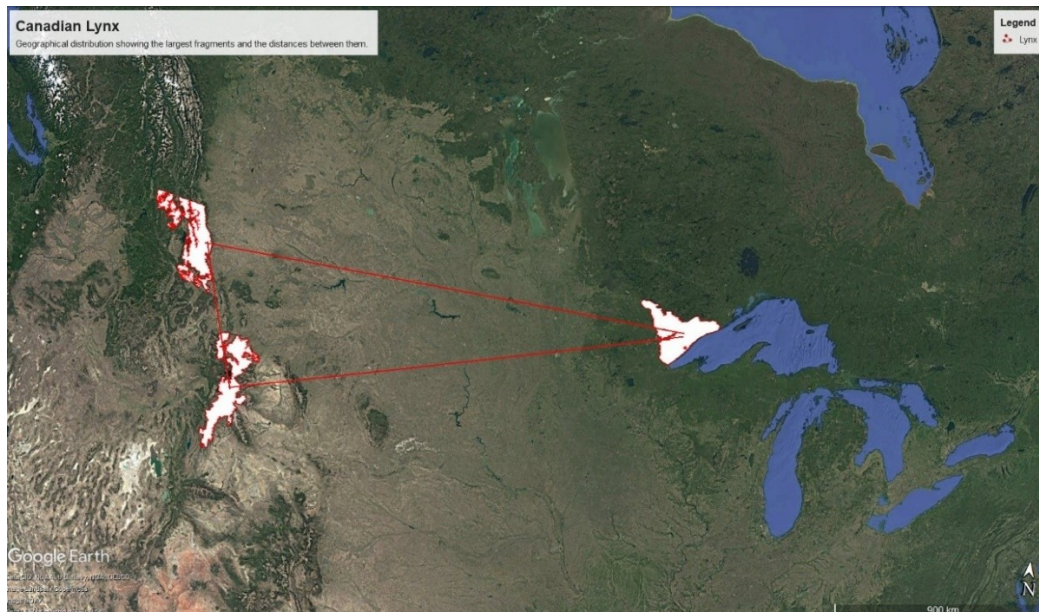
3.2. Bull Trout (*Salvelinus confluentus*): Fish



- i) The regular patch metrics are modified to accommodate the aquatic habitat. From the KML file we can identify 15 major water thresholds with the longest connected waterway being ~250 km and the shortest being ~5 km.
- ii) Dams and diversions disrupt 80 % of the flow, creating fragmentation.

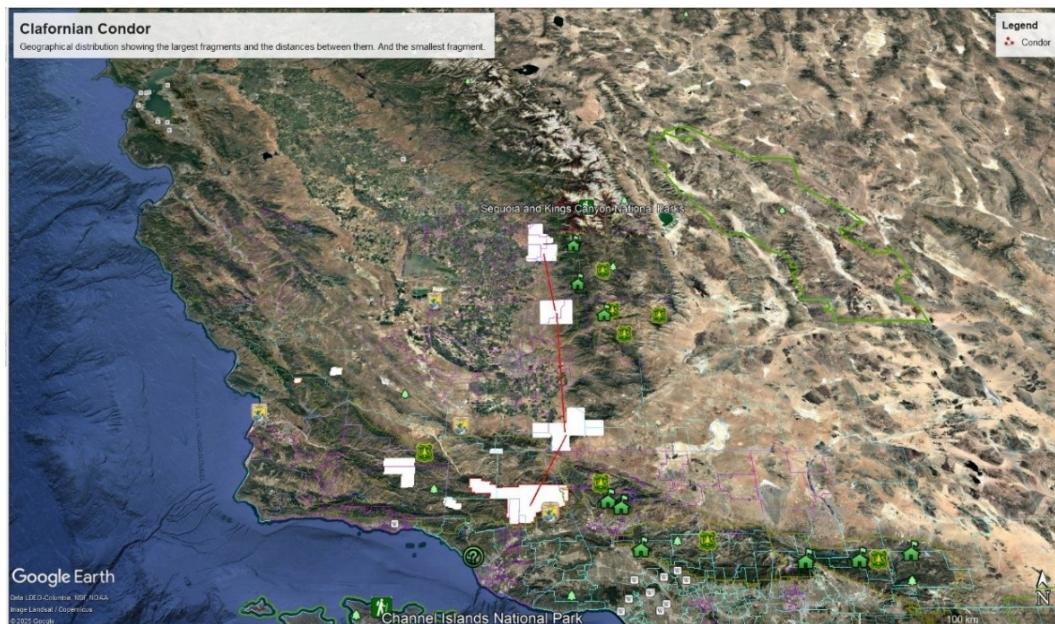
- iii) The river system shows limited connectivity.
- iv) The fragmentation is driven by the barriers and not by any land distances.

3.3. **Canada Lynx** (*Lynx canadensis*): Mammal



- i) Habitat is spread out into 6 clusters, with the largest fragment being 9145 sq.km and the smallest being ~1840 sq.km.
- ii) Most of the fragments are near to each other with the longest distance between fragments being 1615 km.
- iii) The species show low fragmentation with reasonable connectivity.

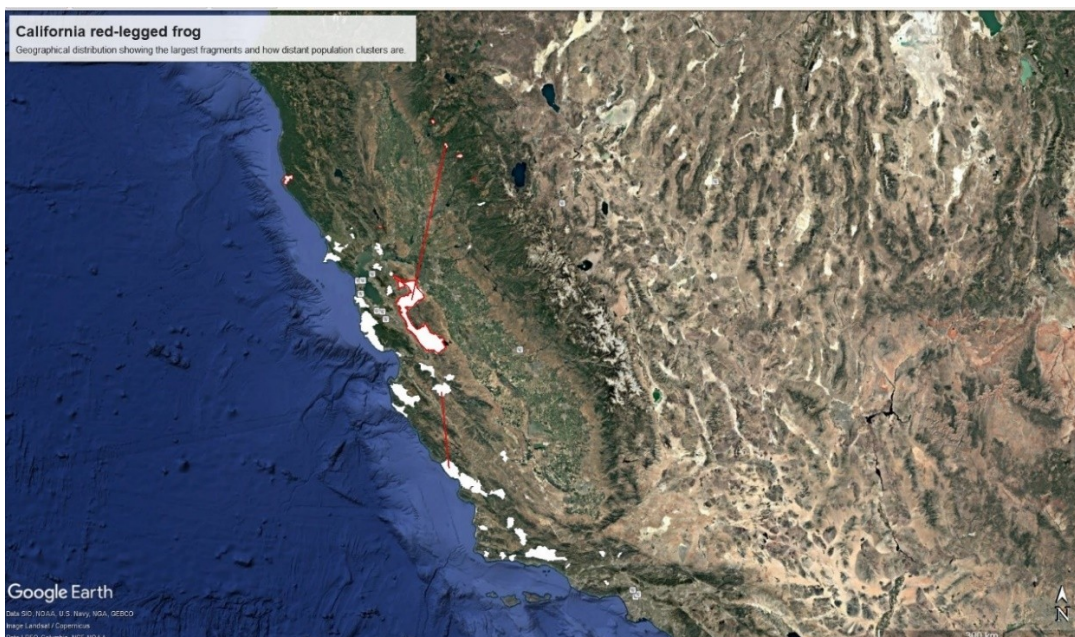
3.4. **California Condor** (*Gymnogyps californianus*): Bird



i) The habitat spread out into 5 clusters, with the largest fragment being ~715 sq.km and the smallest being ~17 km.

ii) The fragments are moderately spread out, but connectivity remains strong due to the high mobility of the species.

3.5. California Red-Legged Frog (*Rana draytonii*): Amphibian

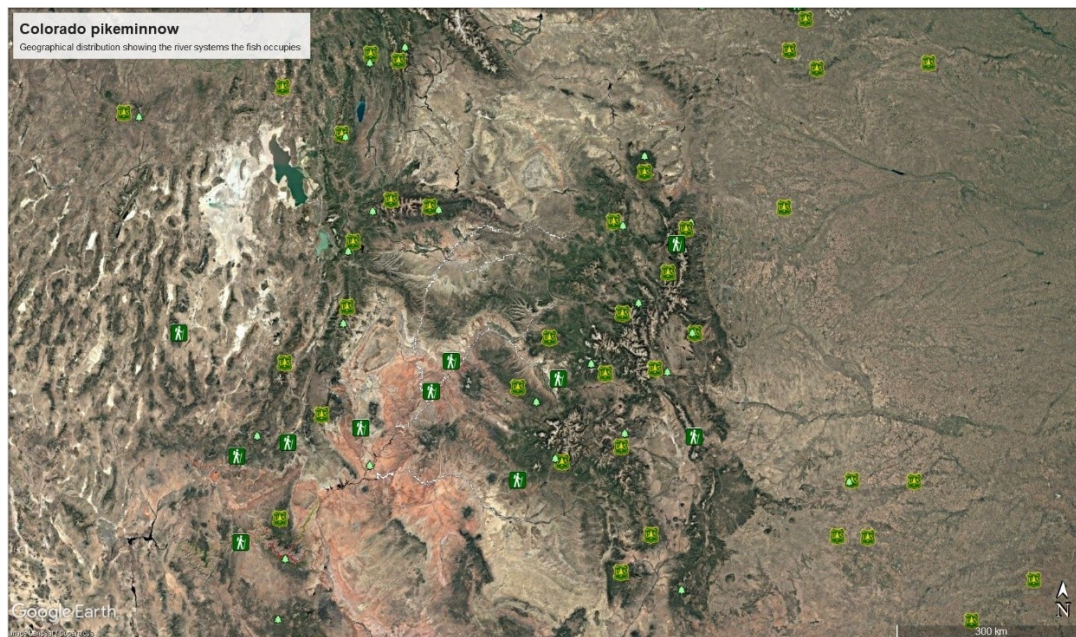


i) From the KML file have identified ~20 clusters, with the largest being 1928 sq.km and the smallest being ~5.3 sq.km.

ii) Most fragments can be grouped into 2 major areas with few isolated outliers in the Rocky Mountains. The maximum distance between two fragments is 212 km.

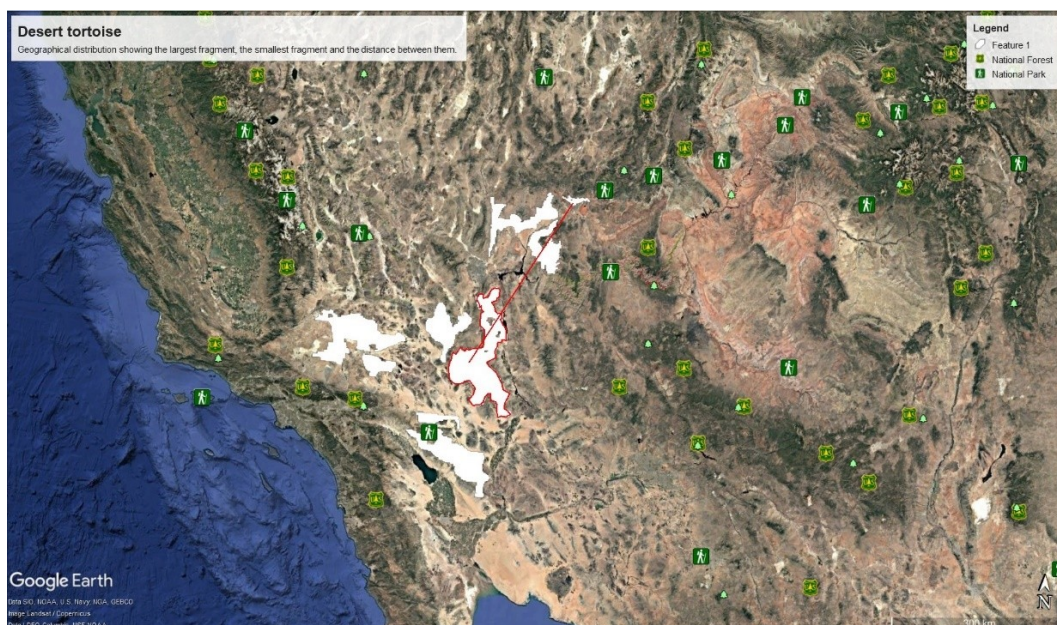
iii) The species exhibits very sever fragmentation due to significant isolation and dependance on the wetland connectivity.

3.6. Colorado Pikeminnow (*Ptychocheilus lucius*): Fish



- i) The regular patch metrics are modified to accommodate the aquatic habitat. From the KML file we can identify 2 major water thresholds. The longest connected waterway being ~170 km and the smallest being ~30 km.
- ii) Heavy Damming and blocks disrupt 100% of the upstream flow for the species.
- iii) The sever fragmentation is driven completely by the barriers.

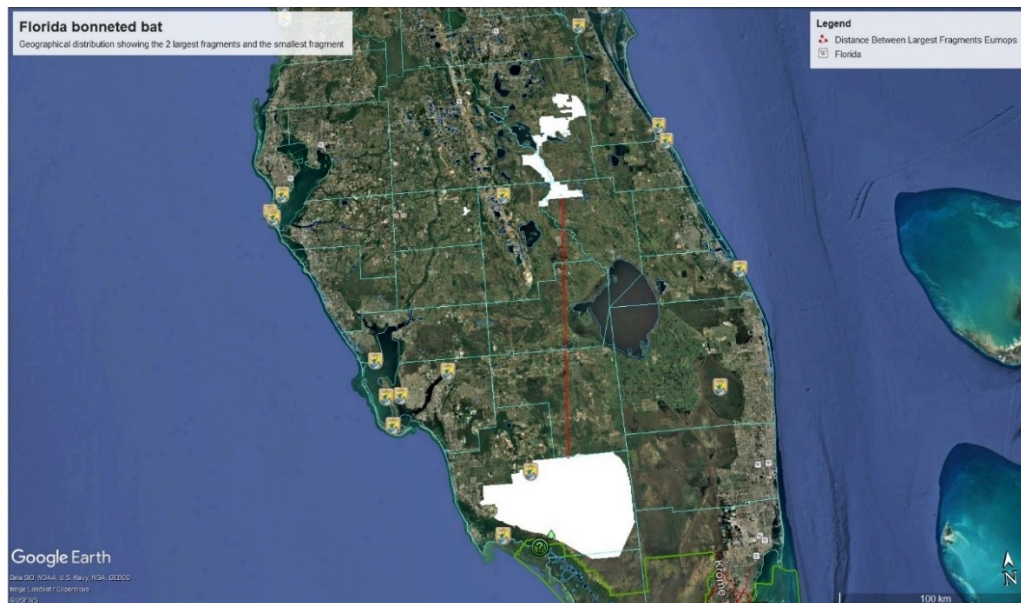
3.7. Desert Tortoise (*Gopherus agassizii*): Reptile



- i) The habitat has 12 major clusters, with the largest fragment being ~7200 sq.km and the smallest fragment being ~240 sq.km.
- ii) The fragments are fairly close to each other but with some highways blocking the connectivity.

iii) The severity of the fragmentation medium-high due to the animals' slow mobility, the harsh weather and the poor connectivity.

8. **Florida Bonneted Bat** (*Eumops floridanus*): Mammal

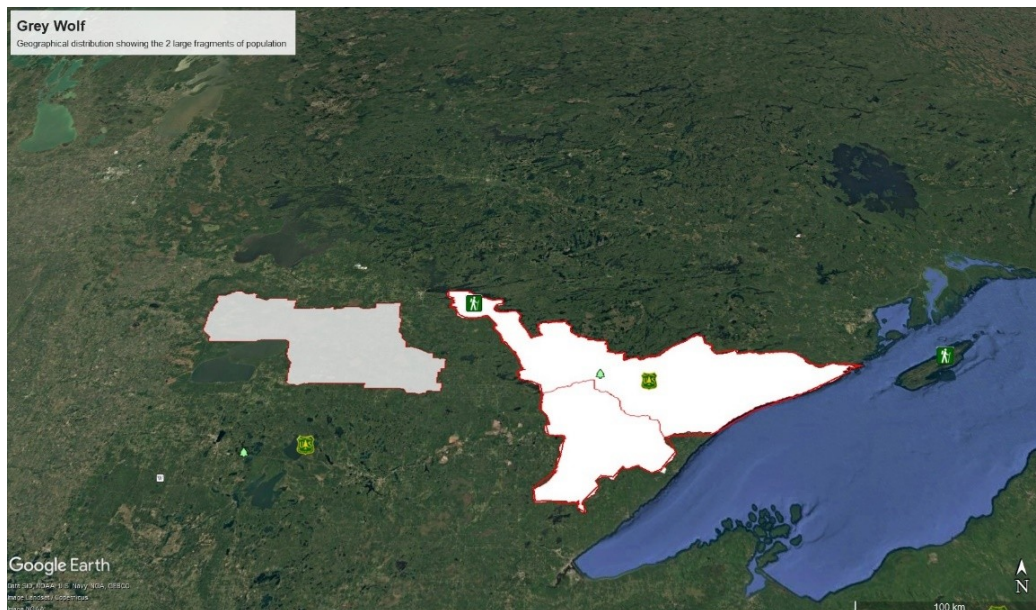


i) This habitat is spread across 7 distinct fragments, with the largest fragment being ~690 sq.km and the smallest being ~10.6 sq.km.

ii) The fragments are spread around an urban area with the longest distance between fragments being ~150km.

iii) The fragmentation severity is moderate as the clusters are spread around an urban area, only slightly limiting the mobility of the animal.

3.9. **Gray Wolf** (*Canis lupus*): Mammal



- i) The habitat is spread across 3 major clusters. The largest being a combination of 2 fragments covering a total of 15247 sq.km and the smaller fragment being 4277 sq.km.
- ii) The distance between the combined large fragment and the smaller fragment is ~250 km.
- iii) Fragmentation severity is low due to the exceptional mobility of the species and large fragments.

3.10. **Haha** (Cyanea Superba): A Hawaiian plant.



- i) The habitat is spread across 4, with the largest being ~18 sq.km and the smallest being ~1.4 sq.km.
- ii) The fragments are spread around mountainous terrain making pollination tricky.
- iii) The severity of fragmentation is extreme due to the endemic nature of the species and the total area occupied is small.

VIII. Temporal Analysis:

1. First, a new work sheet named “Temporal Analysis” is created in which all the columns from “Selected_Species” are carried over.
2. The year of designation for each of the selected 10 species is extracted and placed them in the newly created ‘Designation Year’ column.
2. The years are separated into 3 bins 1960-2000, 2001-2012 and 2013- 2025.
3. Next, I calculated the Average of number of clusters, area of largest fragment, area of smallest fragment, and max separation for each bin.

[illegible]

3.1. Bin 1

- i) It has 4 species.
- ii) Relatively low average number of clusters at 5.5.
- iii) Has the largest average for largest fragments at 5833 sq.km.
- iv) Has the largest average for small fragments at 1128.5 sq.km.
- v) Moderate average max separation at 242.5 km.

3.2. Bin 2

- i) Has 3 species.
- ii) Has highest average number of clusters at 13.
- iii) Has lowest average for largest fragments at 732 sq.km.
- iv) Has lowest average for small fragments at 13.44 sq.km.
- v) Has lowest average max separation at 90.66 km.

3.3. Bin 3

- i) Has 3 species.
- ii) Average number of clusters is 5.67.
- iii) Has a medium average for largest fragments at 3280 sq.km.

iv) Has a medium average for small fragments at 617 sq.km.

v) Has highest average max separation at 628 km.

4. Interpretation of findings:

i) There is a clear decline in the size of the of both large and small fragments from the early period to the middle period and there is some rebound in the latest period.

ii) The max separation distance has dramatically increased over the years, severely increasing the vulnerability of species.

XI. Comparative Analysis:

1. Taxonomic patterns:

i) Mammals show the greatest range of severity ranging from low in Grey Wolves to extreme in Buena Vista Lake ornate shrew.

ii) Fishes consistently show that poor connectivity is due to dams.

iii) Amphibians consistently show high fragmentation with small and isolated fragments.

iv) Animals with flight capabilities can easily overcome the shortcomings of fragmentation with their excellent mobility.

2. Body Size and Mobility correlation:

Large and more mobile species have much larger fragments and lower isolation when compared to the smaller and less mobile species.

3. Habitat Type Comparison:

i) Forest dwelling species usually have contiguous fragments with better connectivity.

ii) Wetland/Marsh dwelling species naturally have a very patchy distribution and are dependent on the availability of water for survival. They are also the most sensitive to human encroachment they can't easily relocate.

iii) Aquatic species are bound to the available water basins and river that connect them. They are already heavily affected by human activity in the form of dams.

iv) Desert dwelling species have to live in remote areas with natural patchiness which is being exacerbated by human developments like highways.

v) Mountain dwelling species have to contend with the topographical features. Human encroachment is very low due to this difficult terrain, so, the animals are less affected by fragmentation.

vi) Island dwelling species are restricted to a very limited area and are very sensitive to any changes.

4. Geographic patterns:

i) On the west coast of North America (Desert Tortoise, California Condor, California Red-Legged Frog) fragmentation is greatly influenced by the mountainous nature of the terrain and the widespread human development restricts the movements of the species.

ii) In the forested regions (Grey Wolf, Canada Lynx) the fragments are large and maintain good connectivity through natural corridors.

iii) In the southern coastal region (Florida Bonneted Bat) medium to high fragmentation can be seen, this is mainly due to intense urbanization.

v) In the pacific islands (Haha) the fragmentation is sever and is mainly driven by its isolation, topography and the rapid human development in the lowlands.

vi) In the River system (Bull Trout, Colorado Pikeminnow) the fragmentation is due to poor connectivity which is mainly because of damming of rivers.

X. Conclusion:

The comparative analysis reveals that fragmentation patterns differ significantly across the different regions, taxonomic groups and habitat types. The severity of fragmentation and its effects upon a species are strongly influenced by the size, mobility and habitat specialization. The most severely fragmented species is the Buena Vista Lake Ornate Shrew with t which possess only tiny fragments which are spread across a large area. The other highly vulnerable species is the Haha plant, which is only found on the small island of O'ahu.

The conservation efforts should focus on protecting the habitat and connectivity of these highly vulnerable species. It should be recognized that each species might require a different conservation approach. The temporal analysis further adds to this, showing that the earlier designated areas are usually much larger when compared to the newer ones. It is clear that over time the loss of habitat and fragmentation has increased rapidly.

References

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